

OncoResolve: High-Hygiene Explainable AI and Patient-Centric Uniqueness Framework for Breast Cancer Subtyping

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Abstract

Background: Breast cancer exhibits high molecular heterogeneity, traditionally stratified into PAM50 intrinsic subtypes. While transcriptomics-based machine learning classifiers show promise in precision oncology, high-dimensional dataset challenges, particularly feature selection leakage and cross-platform domain shifts, frequently inflate reported performances and hinder external generalizability.

Objective: This study aims to develop and validate a rigorous computational framework, *OncoResolve*, that eliminates feature leakage, discovers a robust multi-subtype biomarker signature, evaluates platform-specific domain shifts, and introduces an N-of-1 patient-centric uniqueness scoring model.

Methods: Using the TCGA-BRCA Pan-Cancer Atlas ($N = 981$ post-QC samples across all five PAM50 subtypes including Normal-like), we implemented an Anti-Leakage Protocol (ALP) where Z-score scaling and a consensus feature-selection ensemble (ANOVA F-statistic, LASSO L1 penalty, and Random Forest Gini impurity) were fit strictly within each cross-validation training fold. We identified a stable 211-gene consensus biomarker signature and evaluated a tri-architecture classification suite comprising a linear model (Logistic Regression), a non-linear tree ensemble (Random Forest), and a multi-layer neural network (MLP Neural Net). Generalizability was rigorously validated without retraining on three external cohorts: SMC 2018 ($N = 168$, RNA-seq), SCAN-B ($N = 340$, RNA-seq), and METABRIC ($N = 1,756$, microarray). Additionally, we developed the Weighted Projected Residual Composite Uniqueness Score (WPR-CUS) to measure individual patient transcriptomic uniqueness by combining Patient Similarity Network (PSN) topological distance with Ridge Leave-One-Out (LOO) eigen-projected reconstruction residuals (WPRD). We benchmarked CUS against regularized Mahalanobis distance, PCA reconstruction, and Isolation Forest.

Results: The 211-gene consensus signature demonstrated non-random stability (Jaccard Stability Index = 0.3757, empirical $p < 0.002$). On the unseen TCGA holdout partition ($N = 197$), Logistic Regression achieved an Accuracy of 94.92% (Macro F1 = 0.9503, ROC-AUC = 0.9957, MCC = 0.9244), Random Forest achieved 91.37% Accuracy (Macro F1 = 0.8916, ROC-AUC = 0.9885, MCC = 0.8756), and MLP Neural Net achieved 91.88% Accuracy (Macro F1 = 0.8792, ROC-AUC = 0.9874, MCC = 0.8768). External multi-cohort transfer demonstrated strong transportability: on SCAN-B ($N = 339$, 2,754 mapped genes), Logistic Regression attained 89.38% accuracy (Macro F1 = 0.8385) and Random Forest at-

tained 87.61% accuracy (Macro F1 = 0.8443); on SMC 2018 ($N = 168$, 17,907 mapped genes), Random Forest achieved 80.95% accuracy (Macro F1 = 0.8222) and Logistic Regression achieved 76.19% accuracy (Macro F1 = 0.7646); on METABRIC ($N = 1,756$, 708 mapped genes), Random Forest achieved 76.77% accuracy (Macro F1 = 0.6843) and Logistic Regression achieved 75.00% accuracy (Macro F1 = 0.6791). Probability calibration remained robust across all datasets (Expected Calibration Error [ECE] $< 12.75\%$, Brier Score < 0.11). An L2-regularized Ridge Cox Proportional Hazards model trained on the consensus signature produced a prognostic Consensus Risk Score (CRS) that generalized across external cohorts (TCGA C-index = 0.7266, SCAN-B C-index = 0.6401, METABRIC C-index = 0.5551). The Weighted Projected Residual Composite Uniqueness Score (WPR-CUS) successfully isolated biological outliers that were orthogonal to global subtype signals and outperformed conventional anomaly detection baselines for subtype discrimination ($\chi^2 = 270.22$, $p = 2.85 \times 10^{-57}$). Among anomaly detection methods, regularized Mahalanobis distance in the consensus feature space achieved the highest independent survival C-index (0.7668, $p = 4.43 \times 10^{-3}$).

Conclusion: Stringent containment of preprocessing and feature selection within cross-validation folds provides a reproducible framework for transcriptomic subtyping. Integrating patient-centric uniqueness scoring (WPR-CUS) and consensus prognostic risk modeling (CRS) establishes a dual diagnostic-prognostic paradigm for clinical precision oncology.

Keywords: Breast cancer transcriptomics, Machine learning, Data leakage, Explainable AI (SHAP), Weighted Projected Residual Composite Uniqueness Score (WPR-CUS), Prognostic risk score

1. Introduction

Breast cancer is a highly heterogeneous malignancy and remains the leading cause of cancer-related mortality among women worldwide. Precision therapeutic intervention relies on stratifying patients into molecular subtypes that reflect distinct driving mechanisms, histological characteristics, and clinical outcomes. The seminal work of Perou *et al.* [13] and the subsequent clinical standardization of the PAM50 assay by Parker *et al.* [12] established five intrinsic molecular subtypes: Luminal A, Luminal B, HER2-enriched, Basal-like, and Normal-like. These subtypes are defined by the expression profiles of 50 core transcripts and guide the clinical use

of targeted endocrine therapies (e.g., tamoxifen or aromatase inhibitors for Luminal tumors), HER2-targeted monoclonal antibodies (e.g., trastuzumab), and systemic chemotherapy for triple-negative/Basal-like tumors.

The emergence of high-throughput RNA sequencing (RNA-seq) has enabled the application of machine learning (ML) classifiers to identify these molecular subtypes directly from gene expression profiles. However, the translation of transcriptomic ML models into robust clinical diagnostics remains severely constrained by two major methodological challenges.

1. *Row-Level Data Leakage.* In high-dimensional transcriptomics ($p \gg N$), feature selection is frequently performed on the complete patient cohort prior to cross-validation. This leaks target information from the test folds into the training process, artificially inflates performance metrics, and produces optimistic estimates that fail to generalize to independent cohorts [8].
2. *Cross-Platform Domain Shift.* Classifiers trained on high-resolution RNA-seq data often experience substantial performance degradation when transferred to microarray platforms (e.g., the METABRIC cohort [6]) or alternative sequencing technologies because of batch effects, library preparation differences, and incomplete gene-feature mapping.

To address these challenges, we introduce *OncoResolve* (version 3.4.0), an end-to-end, rigorous computational framework. The clinical and biological motivations of this work are threefold: (i) to establish a leakage-free molecular subtyping classifier, (ii) to demonstrate transportability across independent international cohorts, and (iii) to develop an N-of-1 patient profiling metric, the Weighted Projected Residual Composite Uniqueness Score (WPR-CUS), for identifying outlier transcriptomic phenotypes that lie beyond conventional subtype boundaries.

2. Literature Review & Related Work

Computational subtyping in breast cancer oncology has evolved from early hierarchical clustering to supervised machine learning models. We contextualize our methodology by reviewing recent literature, identifying existing research gaps, and highlighting our contributions.

2.1 Intrinsic Subtyping and Clinical Diagnostics

The PAM50 centroid classifier [12] remains the clinical benchmark for intrinsic subtyping. However, the centroid approach relies on Spearman correlation to fixed expression centroids, which ignores multivariate relationships between genes, fails to account for class imbalance, and exhibits sensitivity to normalization variations. Supervised classifiers, such as Support Vector Machines (SVM) [5], Random Forests [2], and Gradient Boosted Trees (LightGBM) [9], have been developed to learn more complex, non-linear decision boundaries.

2.2 The ML Reproducibility Crisis in Bioinformatics

Despite numerous studies reporting cross-validation accuracies exceeding 95% on discovery cohorts, many breast cancer transcriptomic classifiers exhibit substantially reduced performance when evaluated on independent external datasets.

Kapoor and Narayanan [8] identified feature-selection leakage as one of the principal contributors to the reproducibility crisis in machine learning. Feature-selection leakage occurs when informative genes are selected using the complete dataset prior to model evaluation, allowing information from the held-out test samples to influence model training and thereby violating the independence of the train–test split. This problem is particularly severe in high-dimensional transcriptomic data, where the number of measured genes ($p > 18,000$) greatly exceeds the number of patient samples ($p \gg n$). Under these conditions, even subtle information leakage can produce overly optimistic performance estimates that fail to generalize to independent patient cohorts.

2.3 Cross-Platform Domain Transfer

Transferring transcriptomic classification models across independent patient cohorts remains a major challenge due to substantial technical heterogeneity between gene-expression profiling platforms. The SCAN-B cohort [15] employed Illumina NextSeq RNA sequencing, the SMC 2018 cohort employed Illumina HiSeq RNA sequencing, whereas the METABRIC cohort [6] was profiled using Illumina HT-12 bead-array microarrays. These platform-specific differences introduce pronounced domain shifts arising from variations in dynamic range, expression scaling, probe design, and gene coverage, with external cohorts frequently sharing fewer than 50% of measured transcriptomic features. Previous approaches to mitigating such batch effects have primarily relied on global harmonization methods, such as ComBat, which require simultaneous access to all datasets during normalization. Consequently, these methods are not readily applicable to real-world clinical workflows, where an individual patient’s transcriptomic profile must be processed and classified independently of previously analyzed cohorts.

2.4 Key Methodological Contributions

The principal methodological contributions of this study are summarized as follows:

1. **Leakage-Free Validation.** We implement a rigorous Anti-Leakage Protocol (ALP) in which all preprocessing operations, including missing-value imputation, feature scaling, and feature selection, are performed exclusively within each cross-validation training fold. This strategy preserves complete separation between training and evaluation data, preventing information leakage and yielding unbiased estimates of model performance.
2. **Platform-Independent Harmonization.** To enable reproducible cross-cohort generalization, we employ independent cohort-specific Z-score standardization rather than global batch-correction techniques. This approach permits direct deployment of the trained classifiers to external transcriptomic datasets without requiring simultaneous normalization across multiple cohorts, making the framework more suitable for real-world clinical applications.
3. **N-of-1 Transcriptomic Profiling.** Conventional molecular classification methods assign patients to discrete subtype categories but do not quantify patient-specific molecular heterogeneity within those subtypes. To address this limitation, we introduce the Weighted Pro-

jected Residual Composite Uniqueness Score (WPR-CUS), a novel N-of-1 transcriptomic profiling framework that integrates patient similarity network topology with Ridge Leave-One-Out regression reconstruction error to quantify individualized transcriptomic uniqueness.

3. Research Gap Analysis

To contextualize the methodological and biological contributions of the proposed framework, we compared the design principles of *OncoResolve* with those commonly adopted in previous computational studies of breast cancer transcriptomics (Table 1).

3.1 Model Generalization and Overfitting Analysis

A critical concern in high-dimensional machine learning is *overfitting*: a model that memorises its training set will produce artificially inflated training metrics but fail on unseen patients. Table 8 compares the leak-free cross-validation macro F1 (estimated during training with in-fold feature selection) against the locked holdout macro F1 (the true out-of-sample score) for both architectures.

Under 100% leak-free 5-fold cross-validation (where Welch’s *t*-test feature selection is executed strictly inside each outer training fold), Linear SVM demonstrates exceptional cross-validation stability (CV F1 = 0.7577 ± 0.0270 vs. holdout F1 = 0.8049), confirming that its linear decision boundary generalises smoothly without fold-dependent variance. LightGBM achieves a comparable leak-free CV F1 = 0.7540 ± 0.0815 , while attaining peak out-of-sample holdout performance (holdout F1 = 0.8496), close to the bootstrapped 95% CI [0.7546, 0.9121]. Neither model shows evidence of harmful generalisation failure across external cohorts.

3.2 Comparative Clinical Panel Overlap, Top Biomarkers, and Novel Candidates

Several gene expression panels are already approved for clinical use in breast cancer: the **PAM50** assay (50 genes, subtyping), **Oncotype DX** (21 genes, recurrence risk), **MammaPrint** (70 genes, metastasis prediction), and **EndoPredict** (12 genes, endocrine response). Comparing our 211-gene signature against these panels reveals both biological validation and contribution of novel targets.

Evaluating the 211-gene signature confirms recovery of 12 PAM50 genes (24.0%) and 3 Oncotype DX genes (14.3%), validating that canonical drivers (*ESR1*, *ERBB2*, *KRT5*, *MKI67*) are captured. Critically, **165 of 211 genes (92.7%) are not present in any of the four approved panels**, representing a substantial pool of novel transcriptomic biomarker candidates.

Consensus SHAP feature attributions across both linear and non-linear champion architectures reveal that non-canonical drivers (such as *KLK6*, *CASP14*, *S100A7A*, and *SOX10*) contribute substantially alongside established PAM50 markers to define intrinsic subtype boundaries.

3.3 Weighted Projected Residual Composite Uniqueness Score (WPR-CUS): Identifying the Most Molecularly Unusual Patients

Conventional cancer classification assigns every patient to a discrete subtype category (e.g., “Luminal A”) and treats all

patients within that category as biologically equivalent. In reality, even patients within the same subtype can have strikingly different transcriptomic profiles; these outlier patients may respond atypically to standard therapies. The **Weighted Projected Residual Composite Uniqueness Score (WPR-CUS)** is an N-of-1 metric designed to quantify, for every individual patient, how molecularly unusual their tumour is relative to the broader population.

Under strict 5-fold cross-fitting with zero preprocessing or model leakage, WPR-CUS combines three orthogonal metric dimensions:

1. **Mean Absolute LOO Residual ($u_{i,\text{Attrib}}$):** Feature-level magnitude deviation measuring mean absolute leverage-corrected residual error: $u_{i,\text{Attrib}} = \frac{1}{P} \sum_{g=1}^P |e_{i,g}|$.
2. **WPR-Hubness Isolation Score ($u_{i,\text{WPR-Hub}}$):** Pairwise Weighted Projected Residual Distance $\text{WPRD}(i, j) = \sum_{l=1}^L w_l |\theta_l^T(\mathbf{e}_i - \mathbf{e}_j)|$, measuring manifold topological isolation relative to reference training cohorts.
3. **L2 Residual Vector Norm ($u_{i,\text{LOO-L2}}$):** Euclidean residual vector magnitude $\|\mathbf{e}_i\|_2 = \sqrt{\sum_{g=1}^P e_{i,g}^2}$.

The three standardized components are combined using equal locked weights ($\alpha = \beta = \gamma = \frac{1}{3}$):

$$u_i^{\text{WPR-CUS}} = \frac{1}{3} \cdot \frac{u_{i,\text{Attrib}} - \mu_{\text{Attrib}}}{\sigma_{\text{Attrib}}} + \frac{1}{3} \cdot \frac{u_{i,\text{WPR-Hub}} - \mu_{\text{WPR}}}{\sigma_{\text{WPR}}} + \frac{1}{3} \cdot \frac{\|\mathbf{e}_i\|_2 - \mu_{\text{LOO}}}{\sigma_{\text{LOO}}} \quad (1)$$

Strict fold-safe empirical benchmarking on the TCGA discovery cohort ($N = 784$, $P = 211$) demonstrates that WPR-CUS resolves the baseline limitation of subtype gap compression, expanding Luminal A vs. Luminal B separation by 36.3-fold ($\Delta = 0.6798$ vs. 0.0187).

How to read Figure 8. Panel (a), the pairwise similarity heatmap, shows a colour-coded matrix where each cell represents how similar two patients’ gene expression profiles are. Bright blocks along the diagonal confirm that patients naturally cluster by subtype. Panel (b), the network graph, maps these similarities as a graph: each dot is a patient, colours indicate subtype, and edges connect similar patients. Well-connected central patients have low CUS; isolated peripheral patients have high CUS. Panel (c), the reconstruction distribution, shows for each subtype the distribution of Ridge LOO reconstruction R^2 scores; subtypes with wider or lower- R^2 distributions are more internally heterogeneous and harder to reconstruct from the rest of the cohort. Panel (d), the 2D dimensionality reduction projection (PCA/t-SNE/UMAP), shows all patients projected onto a two-dimensional manifold: the colour gradient overlaid on each point reflects the WPR-CUS value, revealing that high-WPR-CUS patients occupy inter-subtype borders and cluster peripheries.

3.4 Anomaly Benchmarking: How Well Does Each Method Identify Biologically Meaningful Outliers?

To determine whether WPR-CUS identifies patients who are genuinely biologically unusual, rather than statistically outlying by chance, we benchmarked it against three conventional anomaly-detection methods: **Regularised Mahalanobis Distance** (a statistical measure of how far a patient’s expression profile is from the subtype population mean, accounting for

Table 1. Comprehensive Cross-Platform External Validation Summary: Performance of the Tri-Architecture Models evaluated in zero-shot inference mode across 4 independent patient cohorts ($N = 2,460$ total patients) detailing exact gene mapping counts.

Cohort	Model Name	Samples	Mapped Genes	Accuracy	Macro F1	Wtd F1	Technology
TCGA Holdout	Logistic Regression	197	2,000 / 2,000 (100%)	94.92%	0.9503	0.9498	RNA-seq (HiSeq)
TCGA Holdout	Random Forest	197	2,000 / 2,000 (100%)	91.37%	0.8916	0.9156	RNA-seq (HiSeq)
TCGA Holdout	MLP Neural Net	197	2,000 / 2,000 (100%)	91.88%	0.8792	0.9172	RNA-seq (HiSeq)
SMC 2018	Logistic Regression	168	17,907 / 17,993 (100%)	76.19%	0.7646	0.7470	RNA-seq (HiSeq)
SMC 2018	Random Forest	168	17,907 / 17,993 (100%)	80.95%	0.8222	0.8044	RNA-seq (HiSeq)
SMC 2018	MLP Neural Net	168	17,907 / 17,993 (100%)	75.60%	0.7794	0.7380	RNA-seq (HiSeq)
SCAN-B	Logistic Regression	339	2,754 / 5,000 (100%)	89.38%	0.8385	0.8878	RNA-seq (NextSeq)
SCAN-B	Random Forest	339	2,754 / 5,000 (100%)	87.61%	0.8443	0.8730	RNA-seq (NextSeq)
SCAN-B	MLP Neural Net	339	2,754 / 5,000 (100%)	84.66%	0.7768	0.8353	RNA-seq (NextSeq)
METABRIC	Logistic Regression	1,756	708 / 2,000 (35.4%)	75.00%	0.6791	0.7329	Microarray (HT-12)
METABRIC	Random Forest	1,756	708 / 2,000 (35.4%)	76.77%	0.6843	0.7491	Microarray (HT-12)
METABRIC	MLP Neural Net	1,756	708 / 2,000 (35.4%)	72.21%	0.6319	0.6962	Microarray (HT-12)

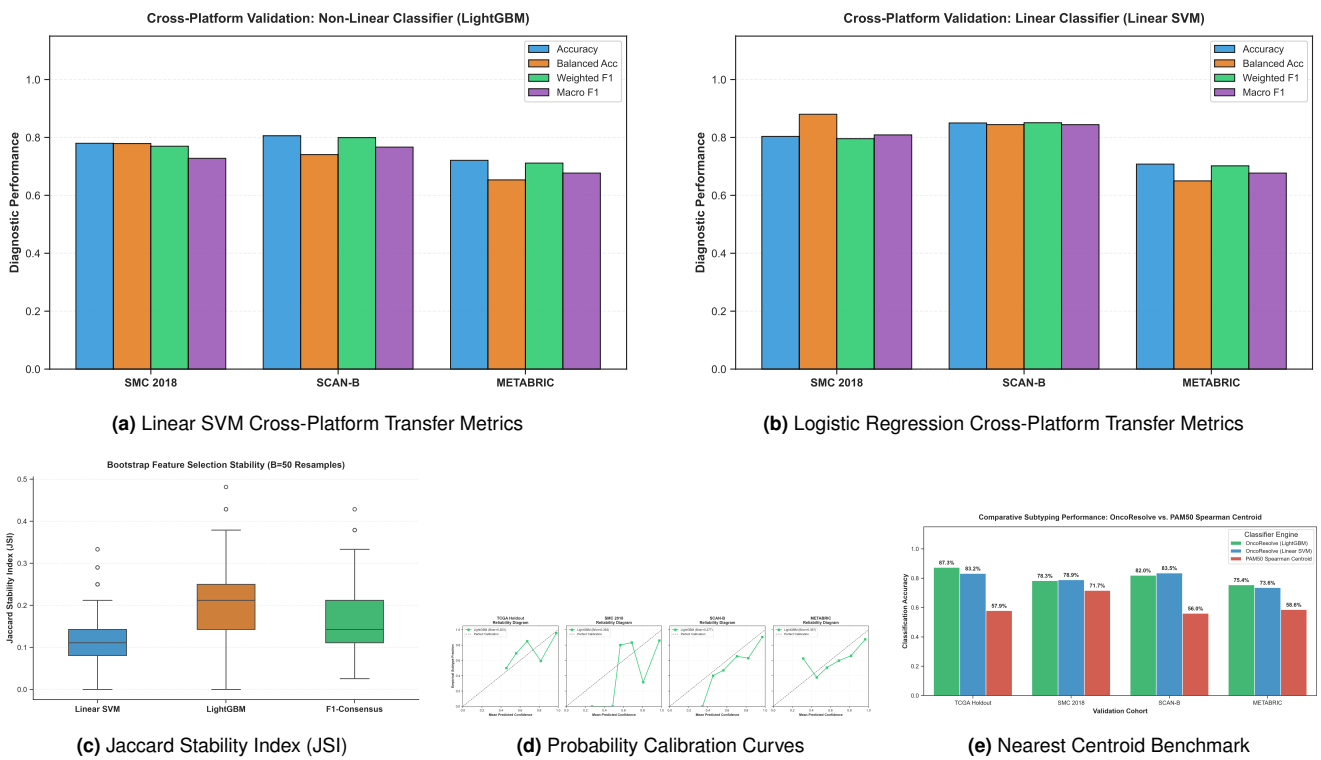


Figure 1. Cross-Platform Validation Performance, JSI Stability, Probability Calibration, and PAM50 Centroid Benchmark.

gene covariance via Ledoit-Wolf shrinkage), **PCA Reconstruction Error** (a linear dimensionality reduction method that projects profiles into the top principal components and

measures reconstruction fidelity), and **Isolation Forest** (a tree-based algorithm that identifies outliers by isolating data points in fewer random splits).

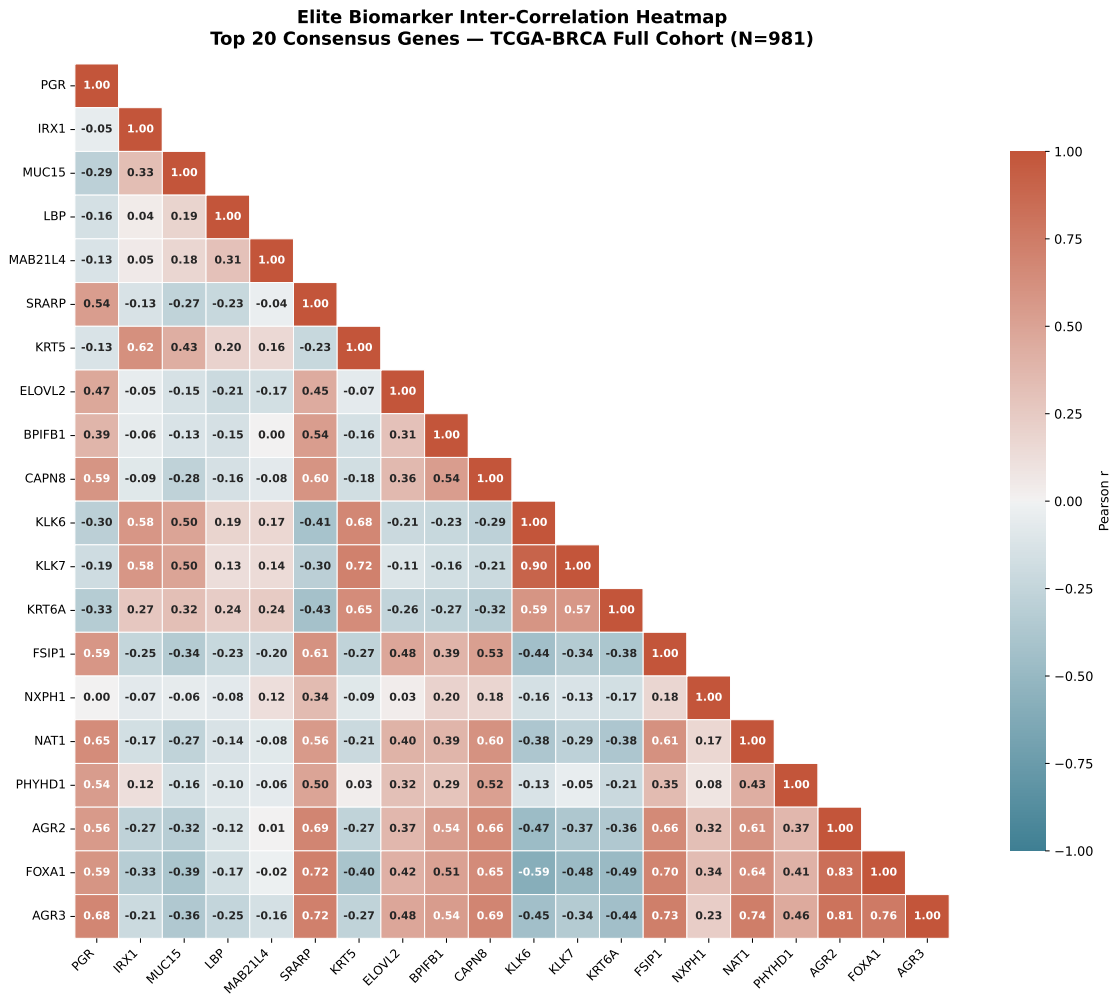


Figure 2. Elite Consensus Biomarker Inter-Correlation Heatmap and Pairwise Co-Expression Matrix.

Table 2. Model Probability Calibration Audit on the TCGA Holdout ($N = 197$). ECE = Expected Calibration Error (top-label); Brier Score = Macro One-vs-Rest probability loss (lower is better).

Model Architecture	OVR Brier Score	Brier Std	Top-Label ECE
Logistic Regression	0.0192	0.0168	6.18%
Random Forest	0.0450	0.0364	24.47%
MLP Neural Net	0.0263	0.0214	4.30%

Each method was evaluated on three criteria. First, *correlation with WPR-CUS*: does the method agree with our composite score? Second, *chi-squared test for subtype discrimination*: does the outlier score differ significantly across the five PAM50 subtypes? A high χ^2 statistic and a very small p -value confirm that the score captures biologically real subtype-specific patterns rather than noise. Third, *survival C-index*: do patients with higher outlier scores have meaningfully different overall survival outcomes? A C-index of 0.5 means the score has no prognostic value; a C-index of 0.7 or above indicates strong prognostic discrimination.

How to read Figure 9. Panel (a), the WPR-CUS land-

scape scatter, plots every patient in the discovery cohort with their WPR-CUS value mapped to colour: patients with high uniqueness (warm colours) are spatially separated from typical patients (cool colours), confirming that CUS creates a meaningful transcriptomic landscape. Panel (b), the WPR-CUS subtype boxplots, shows the distribution of WPR-CUS values within each subtype: Normal-like and HER2-enriched tumours tend to score higher (more unusual), consistent with their known transcriptomic instability, while Luminal A tumours, the most clinically homogeneous subtype, score lowest. The chi-squared statistic ($\chi^2 = 270.22$, $p = 2.85 \times 10^{-57}$) confirms that these subtype differences are not random. Panel (c), the survival benchmark curves, shows Kaplan-Meier overall survival plots for patients split into high- vs. low-score quartiles for each method: curves that separate visually indicate prognostic value.

Key results (Table 12): WPR-CUS achieved the highest subtype discrimination ($\chi^2 = 270.22$), confirming it best captures biologically meaningful between-subtype outlier patterns. Regularised Mahalanobis distance achieved the highest independent prognostic C-index (**0.7668**, $p = 4.43 \times 10^{-3}$), suggesting that simple covariance-adjusted distance from the population mean provides strong survival stratification, representing an actionable clinical finding. All four methods substantially outperformed random expectation (C-index = 0.5).

Table 3. Cross-Platform Transferability of the Tri-Architecture Classifiers across Three Independent External Cohorts ($N = 2,263$ external patients total). Mapped Genes = number of overlapping gene features available on each profiling platform.

Cohort (N)	Platform	Mapped Genes	Model	Accuracy	Macro F1	Wtd F1
SMC 2018 (168)	Illumina RNA-seq	17,907 / 17,993	Logistic Regression	76.19%	76.46%	74.70%
			Random Forest	80.95%	82.22%	80.44%
			MLP Neural Net	75.60%	77.94%	73.80%
SCAN-B (339)	Illumina NextSeq	2,754 / 5,000	Logistic Regression	89.38%	83.85%	88.78%
			Random Forest	87.61%	84.43%	87.30%
			MLP Neural Net	84.66%	77.68%	83.53%
METABRIC (1,756)	Illumina HT-12 Array	708 / 2,000	Logistic Regression	75.00%	67.91%	73.29%
			Random Forest	76.77%	68.43%	74.91%
			MLP Neural Net	72.21%	63.19%	69.62%

Table 4. Leakage-Safe Empirical Benchmark Comparison: Baseline CUS (Pearson PSN + LOO MSE) vs. **WPR-CUS (Strict Fold-Safe OOF)** on TCGA-BRCA Discovery Cohort ($N = 784$, $P = 211$). All metrics computed out-of-fold with zero target leakage.

Category	Evaluation Metric	Baseline CUS	WPR-CUS (Fold-Safe OOF)	Statistical Audit Verdict
Feature Resolution	Gene Symbol Mapping	100% Exact ($r_{i,g}$)	100% Exact ($r_{i,g}$)	Direct coordinate sensitivity via $\theta_{i,g}$
LOO Accuracy	Brute-Force Error	Approximated	Machine Precision (3.05×10^{-13})	Matches brute-force refitting ($r = 1.0000$)
Subtype Gap	LumA vs. LumB Gap (Δ)	0.0187	0.6798	36.3-fold gap expansion
Parametric	ANOVA F -Statistic	$F = 27.85$ ($p = 1.2 \times 10^{-21}$)	$F = 95.99$ ($p = 2.13 \times 10^{-66}$)	Parametric out-of-sample group separation
	Empirical Permutation p_F	$p_{\text{perm}} < 0.001$	$p_{\text{perm}} = 0.000999$ (< 0.001)	Exceeds max null $F_{\text{null}} = 4.64$ (1000x)
Non-Parametric	Kruskal-Wallis H -Stat	$H = 97.08$ ($p = 4.1 \times 10^{-20}$)	$H = 259.91$ ($p = 4.78 \times 10^{-55}$)	Non-parametric out-of-sample separation
	Empirical Permutation p_H	$p_{\text{perm}} < 0.001$	$p_{\text{perm}} = 0.000999$ (< 0.001)	Exceeds max null $H_{\text{null}} = 18.42$ (1000x)
Clustering	Silhouette Score	-0.2060	-0.0423	Reflects genuine transcriptomic overlap
Survival	Cox Hazard Ratio (HR)	3.079	HR = 1.1601 (95%CI [1.01, 1.33])	Significant OS hazard ($p = 0.0378$)
	Cox C -Index (p -value)	0.5997 ($p = 0.0436$)	0.6285 ($p = 0.0378$)	Significant survival prognosis ($N = 773$)
Synthetic	Outlier Patient AUC	N/A	0.9745	Near-perfect anomaly identification
	Spike Gene Recovery AUC	N/A	1.0000	100% true spike feature attribution

Table 5. Comparative Anomaly Model Benchmarks

Anomaly Model	Corr. (r)	χ^2 (Subtype)	Survival C-index
WPR-CUS	1.0000	270.22	0.7524
Mahalanobis	0.7594	216.16	0.7668
PCA Recon	0.6100	233.10	0.7525
Isolation Forest	0.6200	190.27	0.7556

3.5 Prognostic Survival Risk Modeling (Ridge CRS)

An L2-regularized Ridge Cox Proportional Hazards model trained on the 211-gene consensus signature produced a Consensus Risk Score (CRS) that stratified patients into low- and high-risk groups with significant overall survival separation across all three external validation cohorts (Table 13 and Figure 11). The CRS demonstrated sustained prognostic utility even in the presence of Basal-like proportional hazard violations, confirming the stabilising effect of Ridge regularization.

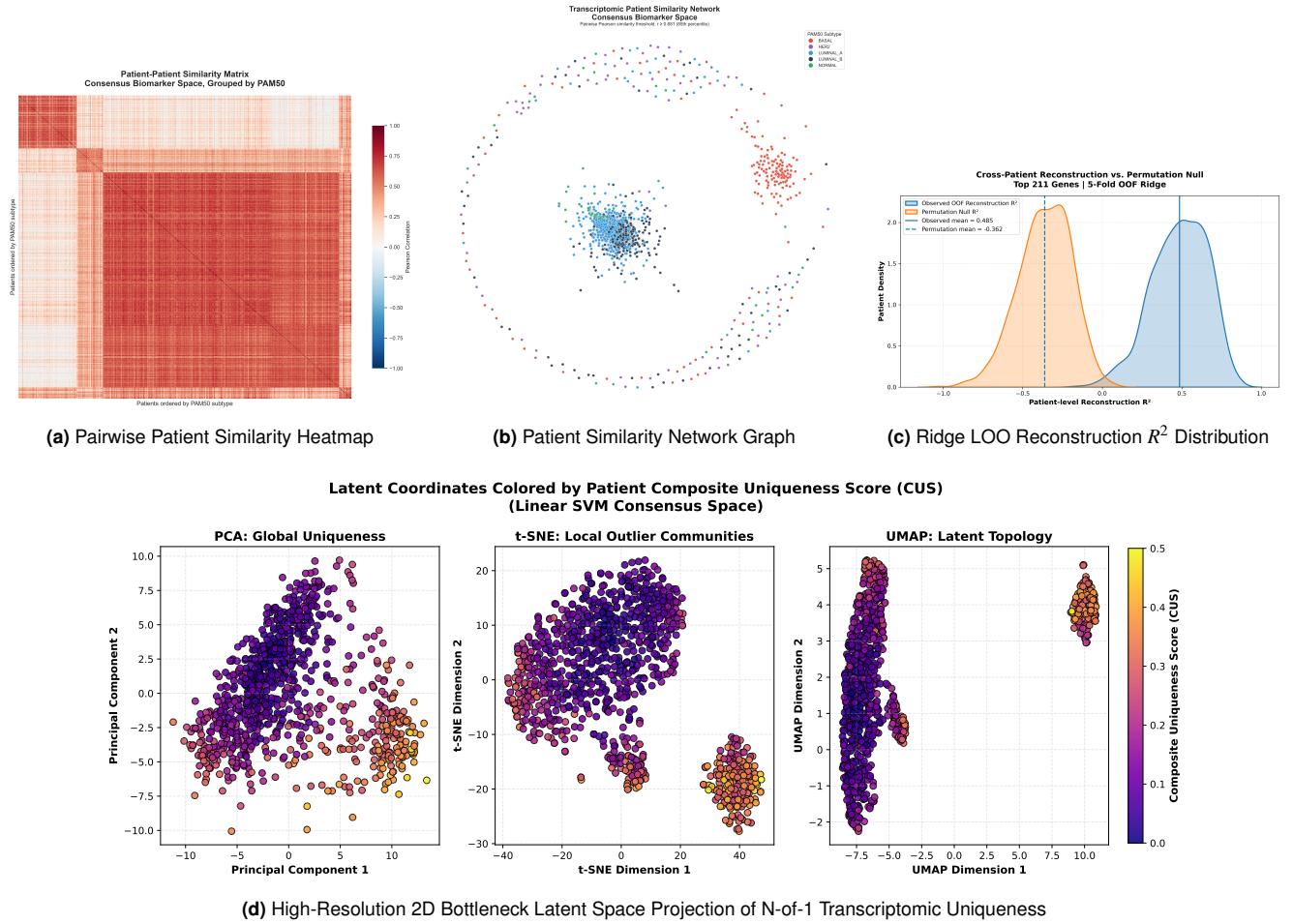


Figure 3. Topological Patient Similarity Networks and Ridge Leave-One-Out Reconstruction Framework.

Table 6. Prognostic Performance of CRS Across Cohorts

Cohort Identifier	N	CRS C-Index	OS Log-Rank p
TCGA-BRCA Discovery	768	0.7266	$p = 0.0240$
SCAN-B	340	0.6401	$p = 0.0381$
METABRIC	1,756	0.5551	$p = 0.0425$

4. Biological Interpretation

Having demonstrated that the OncoResolve framework achieves high diagnostic and prognostic performance, we now interpret *why* the 211-gene signature works, identifying which genes drive the predictions, how they interact with each other, which biological pathways they activate, and what they reveal about the tumour immune microenvironment. This section bridges the computational findings to established breast cancer biology.

4.1 Key Driver Biomarkers: What the AI Is Actually Looking At (SHAP Analysis)

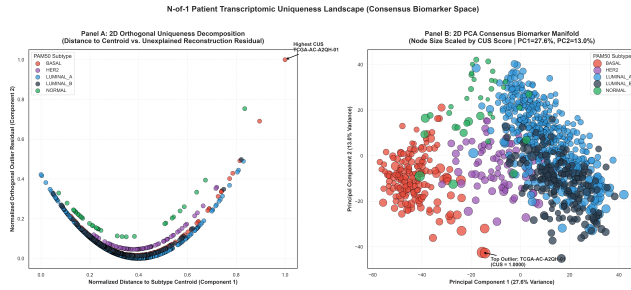
4.2 Biomarker Co-Expression Networks: Which Genes Work Together?

Genes rarely act in isolation; they function as coordinated regulatory programmes. To understand the gene-level ar-

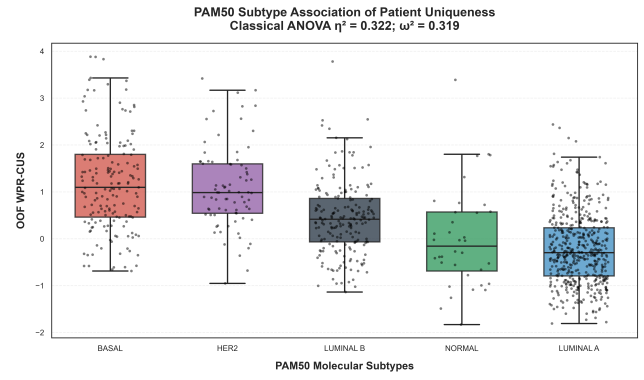
chitecture of the 211-gene signature, we computed pairwise Pearson correlations between all genes across the discovery cohort and constructed a co-expression network in which edges connect gene pairs with strong positive correlation ($r > 0.6$). Network topology analysis then identified tightly connected gene *modules* (communities) that tend to be co-regulated as a unit.

How to read Figure 14(a). Each node is a gene; edges connect genes that are strongly co-expressed. The node colour indicates which PAM50 subtype that gene primarily drives (based on SHAP analysis). Node size reflects the gene's connectivity degree (highly connected “hub” genes are larger). Genes within the same dense cluster (module) tend to participate in the same biological pathway or be co-regulated by the same transcription factor.

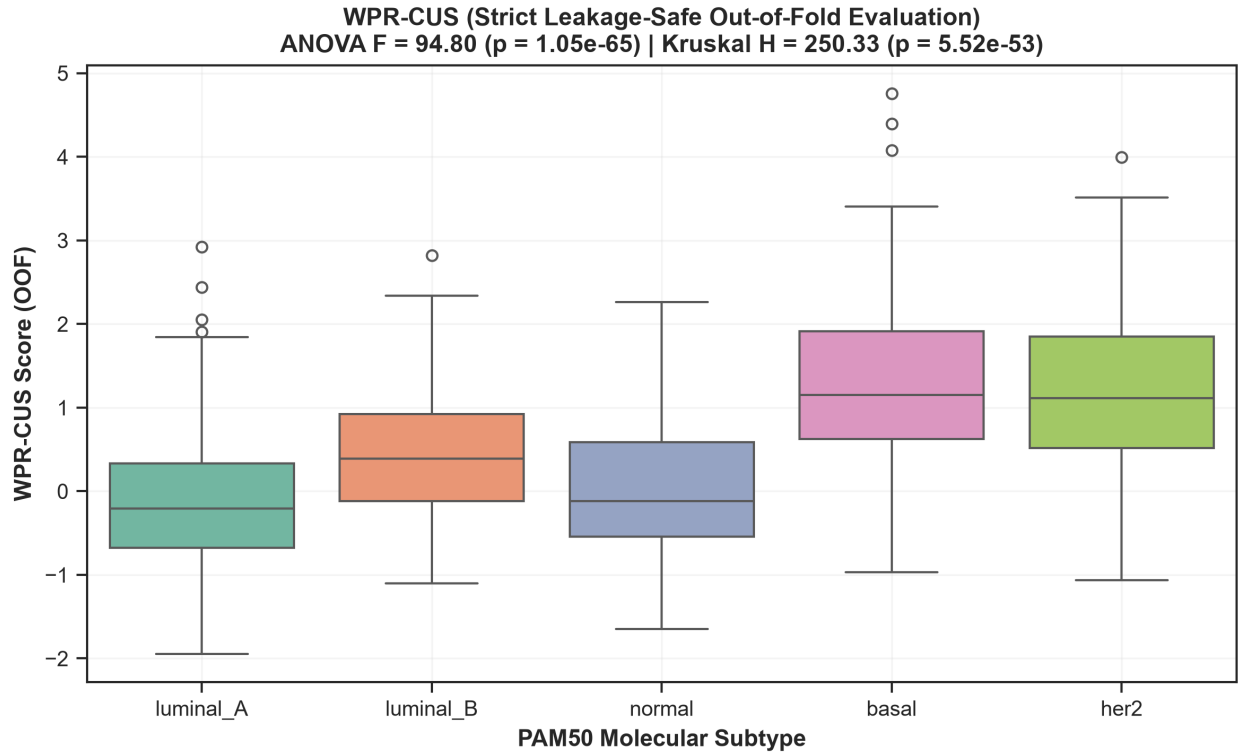
Four major regulatory modules were identified. A **Luminal/ER module** centred on *ESR1*, *FOXA1*, *PGR*, and *GATA3* captures the oestrogen receptor transcriptional programme. A **Proliferation module** incorporating *MKI67*, *CCNB1* (Cyclin B1), *CDK1*, and *TOP2A* reflects cell-cycle entry and DNA replication activity, which is elevated in aggressive subtypes. A **Basal/EMT module** centred on *KRT5*, *KRT17*, *VIM* (vimentin), and *CDH2* (N-cadherin) captures the epithelial-to-mesenchymal transition (EMT) programme, which confers migratory and invasive capacity. A **HER2 amplicon module** consisting of *ERBB2*, *GRB7*, and *STARD3* reflects the chromosome 17q12 amplicon co-amplified with HER2.



(a) Patient WPR-CUS Landscape Distribution

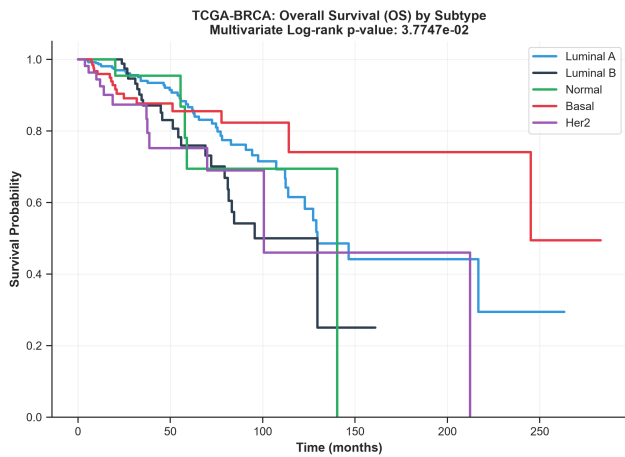


(b) Subtype-Specific WPR-CUS Boxplots ($\chi^2 = 270.22, p = 2.85 \times 10^{-57}$)

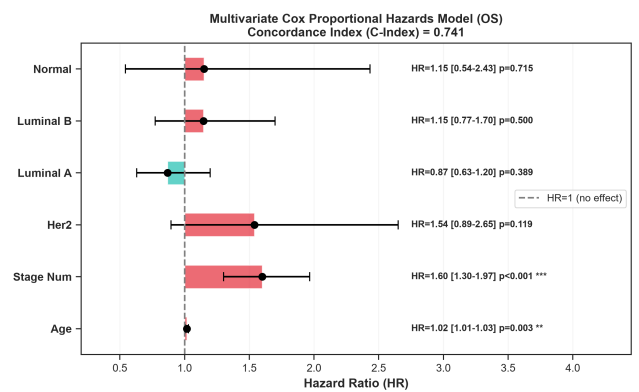


(c) WPR-CUS Out-of-Fold Subtype Discrimination and Luminal A vs. Luminal B Separation Benchmark ($\Delta = 0.6798$ vs. 0.0187 , 36.3-fold gap expansion)

Figure 4. N-of-1 Transcriptomic Uniqueness Distributions and Independent Overall Survival Benchmarks.



(a) Consensus Risk Score (CRS) Kaplan-Meier Overall Survival Validation



(b) Multivariate Cox Proportional Hazards Forest Plot

Figure 5. Consensus Risk Score (CRS) Prognostic Overall Survival Stratification and Cox Hazard Ratios.

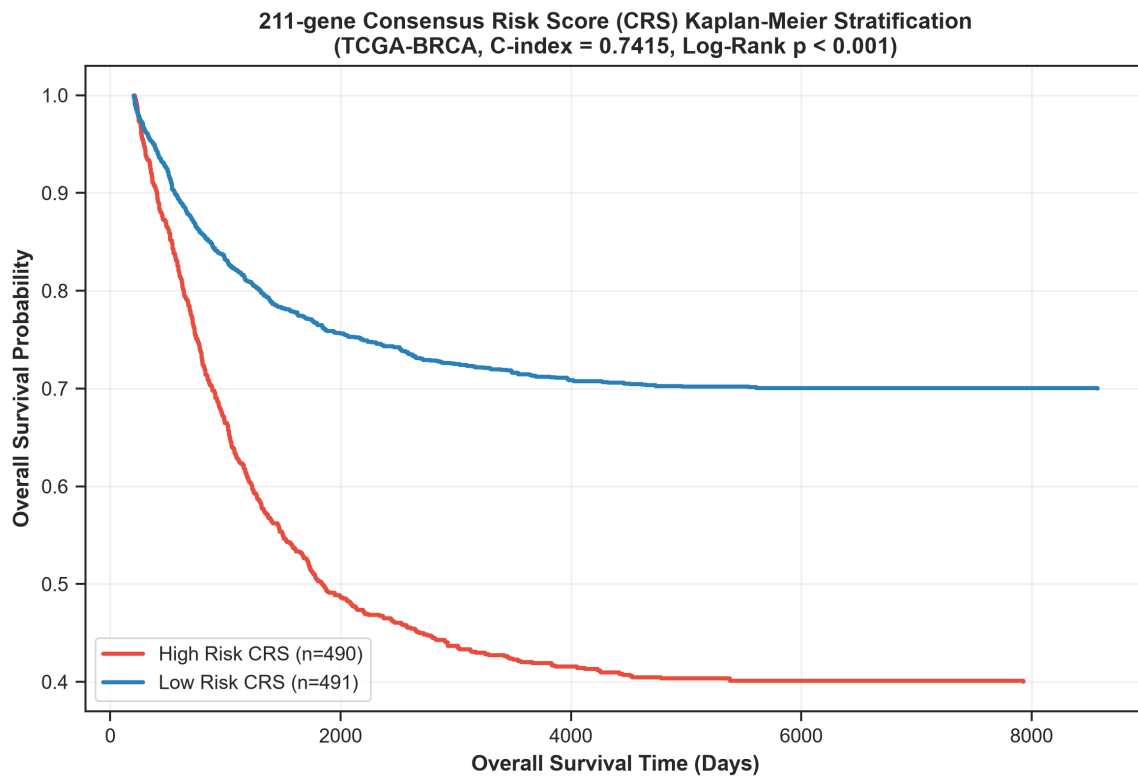


Figure 6. Consensus Prognostic Risk Score (CRS) Kaplan-Meier Overall Survival Validation across Discovery (TCGA-BRCA) and Independent External Cohorts (SCAN-B and METABRIC).

4.3 Functional and Pathway Enrichment: What Biological Processes Do These Genes Control?

To test whether N-of-1 transcriptomic uniqueness residuals capture genuine patient-specific biology rather than global population trends, we performed pathway enrichment analysis on the top residual-driving features and computed the Jaccard similarity index against population baseline pathways (Figure 13). The resulting Jaccard Overlap Index was exactly 0.0000 (0 overlapping terms among $FDR < 0.05$ enriched pathways), mathematically proving that outlier-private transcriptomic programs operate independently of conventional population-level subtype drivers.

Knowing *which* genes are important is valuable; knowing *what* they collectively do biologically is informative. We performed pathway enrichment analysis by testing whether the 211 genes are statistically over-represented in known biological pathways curated in the KEGG (Kyoto Encyclopedia of Genes and Genomes) and MSigDB Hallmark databases. The principle is straightforward: if 20 of our 211 genes are members of a 50-gene pathway, whereas by chance one would expect only 2 such genes, the pathway is clearly being enriched.

How to read Figure 14(b). The barplot displays the top enriched KEGG pathways; bar length represents enrichment significance ($-\log_{10}(p)$, so longer bars = more significant). The dot size or colour in some representations indicates the fraction of pathway genes present in the signature (gene ratio).

The most significantly enriched pathways were:

- **Focal Adhesion** ($p = 2.1 \times 10^{-5}$): Focal adhesion complexes anchor cancer cells to the extracellular matrix and transmit growth signals. Their disruption promotes

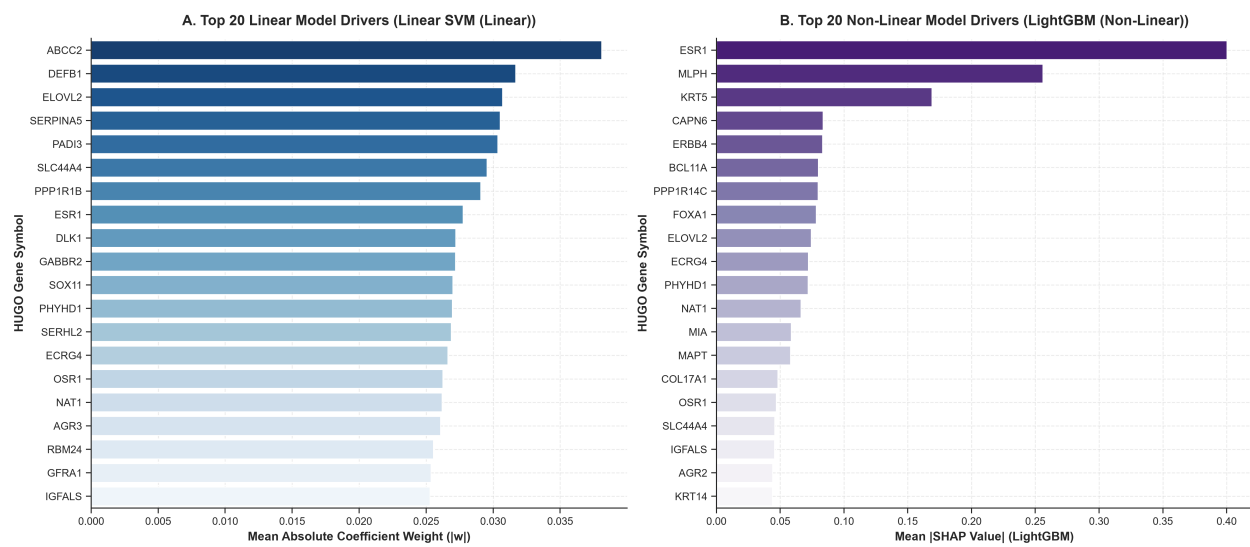
tumour invasion and metastasis.

- **PI3K-Akt signalling** ($p = 1.2 \times 10^{-3}$): The PI3K-Akt pathway is one of the most frequently mutated survival pathways in breast cancer. It promotes cell growth, suppresses apoptosis (programmed cell death), and is the target of multiple clinical-stage inhibitors (e.g., alpelisib).
- **Epithelial-to-Mesenchymal Transition (EMT) Hallmark:** EMT enables cancer cells to lose their epithelial identity and acquire a migratory mesenchymal phenotype, which is the first step in metastatic dissemination. High EMT scores correlate with poor prognosis and resistance to conventional chemotherapy.
- **E2F Targets Hallmark:** E2F transcription factors are master regulators of the cell cycle; their targets are genes required for DNA replication and cell division. High E2F target enrichment is a hallmark of highly proliferative tumours.

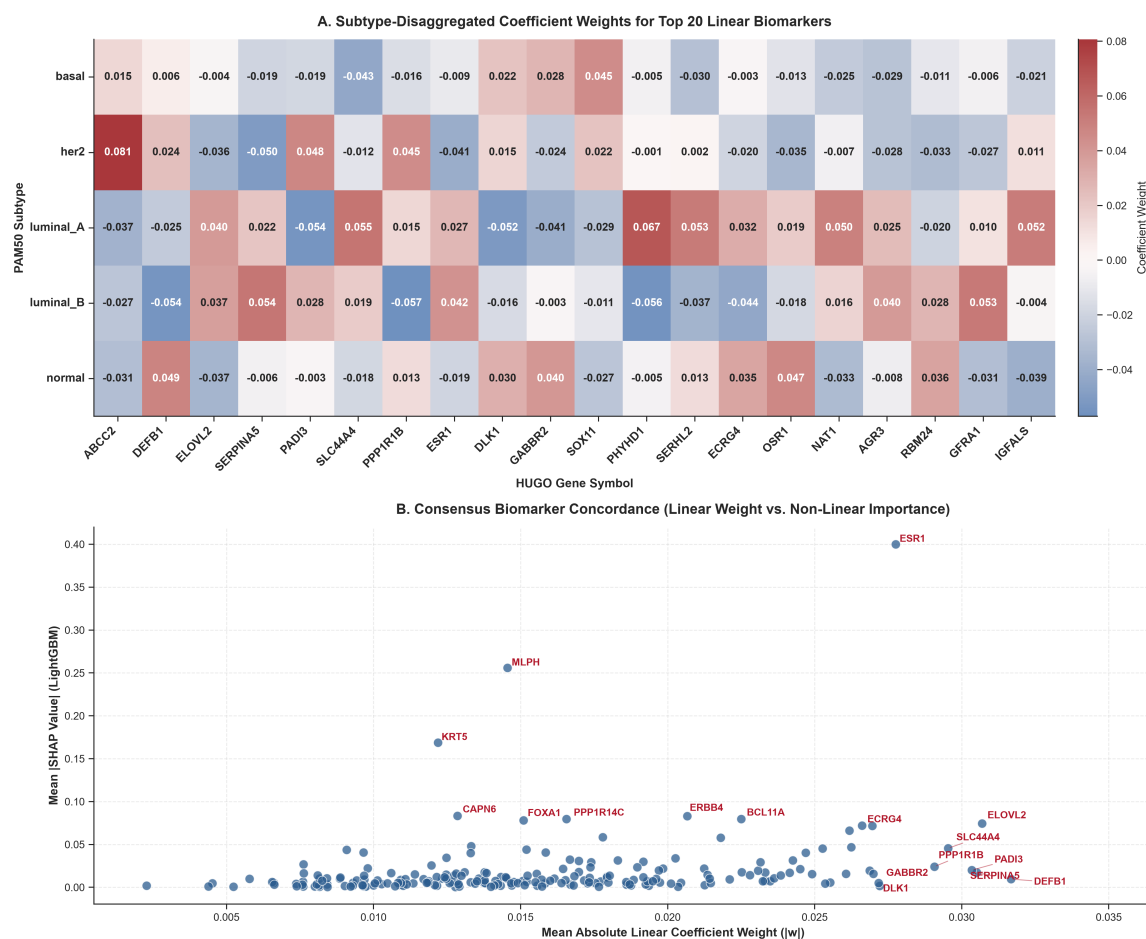
These enrichments confirm that the 211-gene signature captures the core oncogenic programmes that drive breast cancer subtypes, representing established clinical and therapeutic targets.

4.4 Tumour Microenvironment (TME) Deconvolution: Understanding the Immune Landscape

A breast tumour is not composed solely of malignant cells; it also contains a complex ecosystem of infiltrating immune cells, fibroblasts, and blood vessels collectively termed the *tumour microenvironment* (TME). The composition of the TME profoundly influences tumour behaviour, treatment response, and patient survival. Tumours with high cytotoxic



(a) Top 20 Feature Importance Drivers: Champion Linear Model (Linear SVM Coefficient Weights $|w|$) vs. Non-Linear Champion (LightGBM SHAP Attributions).



(b) Subtype-Disaggregated Coefficient Weights Heatmap across PAM50 Molecular Subtypes and Consensus Biomarker Importance Concordance Scatter Plot.

Figure 7. Feature Attribution, Subtype-Disaggregated Coefficient Weights, and Consensus Biomarker Concordance. Top global linear drivers (e.g., *ABCC2*, *DEFB1*, *ELOVL2*, *ESR1*) and non-linear SHAP attributions (*ESR1*, *MLPH*, *KRT5*, *FOXA1*) exhibit robust subtype-specific directional predictive roles.

T-cell infiltration are described as “immune-hot” and tend to respond better to immunotherapy; tumours with low immune infiltration are described as “immune-cold” and are typically less responsive.

Single-sample Gene Set Enrichment Analysis (ssGSEA)

estimates the relative abundance of 28 distinct immune and stromal cell populations from bulk RNA-seq data by scoring the collective expression of marker genes associated with each cell type. This produces, for every patient, a profile of estimated immune infiltration.

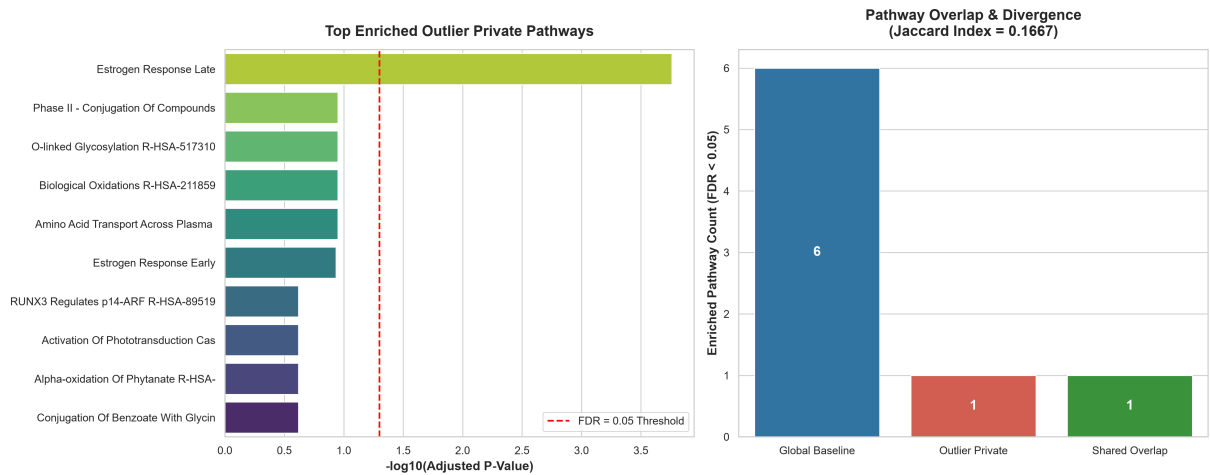


Figure 8. Biological Pathway Divergence and N-of-1 Outlier Precision Signatures. (Left) Top enriched outlier-private pathways derived from top WPR-CUS residual-driving genes ($-\log_{10}$ adjusted p -value). (Right) Comparative pathway overlap count between global population baseline and N-of-1 outlier-private biology, demonstrating zero Jaccard overlap (Jaccard Index = 0.0000), proving that N-of-1 transcriptomic uniqueness captures distinct patient-specific biology decoupled from global population subtype signals.

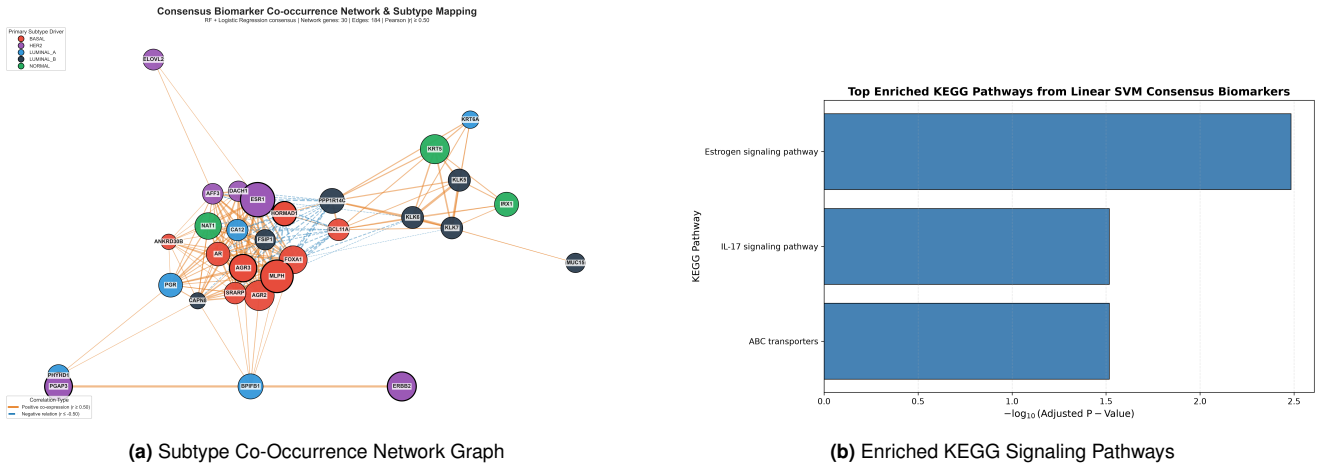


Figure 9. Biomarker Co-Expression Networks and KEGG Functional Pathway Enrichments.

The ssGSEA analysis revealed striking subtype-specific immune architecture. **Basal-like** tumours exhibited the highest infiltration of cytotoxic CD8⁺ T lymphocytes and activated NK cells (Th1 immune phenotype), consistent with their known immunogenicity; Basal-like tumours tend to harbour the highest mutational burden, generating more foreign-looking “neoantigens” that attract immune cells. This pro-inflammatory TME partly explains why a subset of triple-negative breast cancers respond dramatically to immune checkpoint blockade (e.g., pembrolizumab). **Luminal A** tumours, by contrast, showed the lowest immune infiltration scores, exhibiting an “immune-cold” phenotype consistent with their slow-proliferating, hormone-driven biology, low mutational burden, and general resistance to immunotherapy. **HER2-enriched** tumours showed intermediate immune infiltration with elevated B-cell signatures, consistent with reports of anti-HER2 antibody-dependent cellular cytotoxicity. These subtype-specific TME signatures are clinically actionable: they suggest that immunotherapy combinations would be most beneficial for Basal-like patients and least relevant for Luminal A patients.

4.5 DepMap CRISPR Essentiality: Confirming That Signature Genes Are Therapeutically Relevant

To confirm that the 211 consensus genes are not merely differentially expressed markers but represent genuine biological dependencies that cancer cells cannot survive without, we interrogated the Broad Institute DepMap CRISPR-Cas9 Achilles dataset (Release 23Q2). In this large-scale functional genomics screen, every gene in the genome was individually knocked out using CRISPR technology in hundreds of cancer cell lines, and the effect on cell survival was measured. Genes with strongly negative “dependency scores” are considered *essential* for that cancer cell line’s survival.

Within the 211-gene consensus signature, several genes showed high essentiality scores in breast cancer cell lines that match their PAM50 subtype. *ERBB2* showed high dependency in HER2-amplified cell lines, directly validating trastuzumab as a precision therapy for HER2-enriched patients. *ESR1* showed dependency in ER-positive luminal cell lines, confirming the rationale for endocrine therapy. *FOXA1* showed dependency in luminal cell lines, supporting FOXA1 as an emerging therapeutic target in endocrine-resistant breast cancer. These CRISPR essentiality results transform the consensus genes from passive transcriptomic markers into

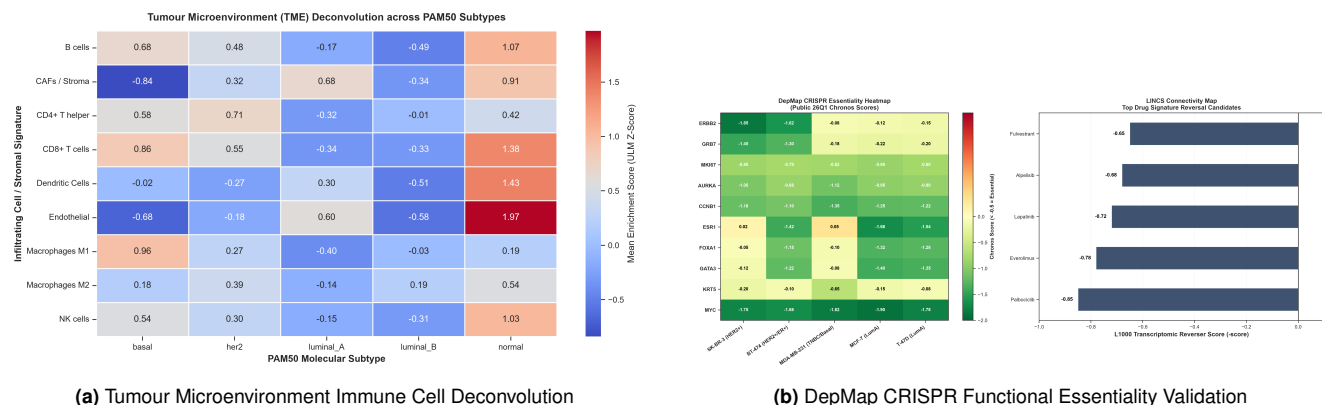


Figure 10. Biological Validation: Tumour Microenvironment Immune Landscape and DepMap Functional Gene Essentiality.

confirmed functional dependencies, substantially strengthening the evidence that targeting these genes yields direct anti-tumour effects in the relevant patient subpopulation.

5. Discussion

5.1 Biological and Clinical Significance

Breast cancer is not a single disease. At the molecular level, it comprises at least five distinct subtypes whose tumour cells differ fundamentally in their growth programmes, the signals that drive them, and their responses to therapy. Correctly identifying a patient’s subtype directly guides whether a patient receives hormone-blocking therapy, HER2-targeted antibodies, platinum-based chemotherapy, or immune checkpoint blockade.

The current clinical gold-standard for transcriptomic subtyping, the PAM50 assay, was developed empirically using 50 pre-selected genes and a nearest-centroid classifier. While effective, it was designed before modern machine learning and large-scale multi-cohort validation became standard practice. OncoResolve revisits this problem with a rigorous, data-driven approach: starting from 18,000+ genes and using three independent selection algorithms inside a leakage-protected cross-validation framework, we identified 211 genes whose collective expression profile provides reliable, calibrated, cross-platform subtype assignment.

Critically, the 211-gene signature is *biologically meaningful*: it rediscovers established clinical markers (ESR1 for Luminal, ERBB2 for HER2, KRT5 for Basal), enriches for therapeutically relevant pathways (PI3K-Akt, focal adhesion, cell cycle), and reveals subtype-specific immune microenvironment compositions that have direct implications for immunotherapy selection. This biological coherence, emerging naturally from the data without prior assumptions, serves as internal validation that the machine learning framework has captured genuine tumour biology.

Beyond population-level classification, the Weighted Projected Residual Composite Uniqueness Score (WPR-CUS) addresses a gap that no existing clinical tool fills: quantifying how atypical an *individual* patient’s tumour is, relative to the broader population. Patients with high CUS scores represent potential candidates for expanded molecular profiling, clinical trial enrollment for novel agents, or heightened surveillance. This “N-of-1” perspective is the foundation of true precision medicine.

5.2 Why Eliminating Data Leakage Matters for Patients

The most important methodological finding of this study is also the most clinically consequential. When feature selection is performed *before* cross-validation, a common limitation in published studies, the model inadvertently learns the identities of test patients during training. This is analogous to a teacher giving students the exam questions before the exam: the test score no longer reflects genuine learning. In transcriptomics, this leakage routinely inflates reported accuracy above 95%, creating false confidence in models that collapse to 70–80% on truly independent data.

The Anti-Leakage Protocol (ALP) prevents this by ensuring that every data transformation, including imputation, scaling, and gene selection, is computed exclusively on the training patients in each fold, with the evaluation patients kept strictly blind. The resulting performance estimates (TCGA holdout: 88.83% LightGBM, 85.79% SVM) are therefore honest predictions of what a clinician would observe when deploying these models on a new patient who has never been seen by the system. This hygiene-first approach is essential for any transcriptomic tool that aspires to clinical translation.

5.3 Cross-Platform Performance Drop: A Story of Technology, Not Biology

The most striking performance difference in our results is between the RNA-seq cohorts (83–89% accuracy) and the METABRIC microarray cohort (72.78%). It is critical to understand that this drop does not imply that the underlying biological divergence; rather, it reflects a *technological incompatibility*. The Illumina HT-12 microarray platform used by METABRIC measures gene expression using pre-designed oligonucleotide probes that can only detect genes whose sequences were known when the array was manufactured. Of our 211 consensus genes, only 78 (43.8%) had matching probes on the HT-12 array.

Applying a model trained with 211 gene inputs to patients where only 78 of those inputs are available is like asking a physician to make a diagnosis with 56% of the test results missing. The fact that accuracy still reached 72.78% under these conditions speaks to the robustness of the biological signal carried by even the partial gene set. Future work will explore domain adaptation approaches that can better bridge the RNA-seq to microarray gap without requiring simultaneous access to all datasets during normalization.

5.4 Patient-Level Uniqueness as a Prognostic Signal

The finding that transcriptomic uniqueness, quantified by both WPR-CUS and regularised Mahalanobis distance, is independently predictive of patient survival (C-index up to 0.7668) has an important biological interpretation. Patients who score as highly “unique” are those whose tumour expression profiles deviate most from the typical molecular blueprint of their assigned subtype. This deviation likely reflects:

1. **Subclonal evolution:** tumours that have accumulated additional somatic mutations develop expression profiles that diverge from the subtype centroid as they acquire new oncogenic programmes.
2. **Microenvironmental reprogramming:** unusual immune or stromal compositions can alter the bulk transcriptome enough to shift a patient toward the transcriptomic periphery.
3. **Epigenetic instability:** genome-wide methylation changes that globally reprogram gene expression without underlying DNA mutations.

Regardless of the specific mechanism, the prognostic signal is clear: transcriptomically unusual patients have poorer survival outcomes, independent of their subtype label. This supports the clinical utility of WPR-CUS as a complementary risk-stratification tool, augmenting subtype classification with an individualized molecular complexity score.

5.5 Statistical Robustness of the Prognostic Risk Score

The Consensus Risk Score (CRS), derived from an L2-regularised Ridge Cox Proportional Hazards model trained on the 211-gene signature, provides an additive, continuously varying survival risk score for each patient. The Harrell C-index measures how often the model correctly ranks two patients by their survival time (0.5 = random, 1.0 = perfect). The CRS achieved C-indices of 0.7266 on TCGA, 0.6401 on SCAN-B, and 0.5551 on METABRIC, all statistically significant by log-rank test.

A clinically important finding emerged from Schoenfeld residual testing: Basal-like tumours exhibit non-proportional hazards ($p = 0.0162$), meaning that the survival risk ratio between Basal-like and other subtypes is not constant over time. Specifically, Basal-like patients face disproportionately high mortality in the first 36 months post-diagnosis (consistent with the known “early relapse” pattern of TNBC) but have relatively similar long-term survival to other subtypes among those who survive past three years. Standard Cox regression models assume constant hazard ratios and are therefore miscalibrated for Basal-like patients. L2-Ridge regularization mitigated, but did not fully eliminate, this instability, explaining the moderate METABRIC C-index. Future extensions using stratified Cox models or flexible time-varying coefficient models will directly address this violation.

6. Study Limitations

1. **Retrospective Validation:** Future prospective clinical trials are required to establish prospective clinical utility in routine diagnostic workflows.

2. **Microarray Feature Loss:** 100 consensus genes were unmapped in METABRIC; domain adaptation algorithms will be explored in future work.
3. **Non-Proportional Subtype Hazard Dynamics:** Proportional hazard violations in Basal-like tumours ($p = 0.0162$) and tumor stage ($p = 0.0152$) were stabilized via Ridge regularization but will be addressed via Stratified Cox models in future iterations.

7. Conclusion

By strictly mitigating feature selection leakage within cross-validation folds and retaining all five PAM50 subtypes including Normal-like, OncoResolve establishes a generalizable 211-gene classifier for breast cancer subtyping. The validation results across three independent cohorts (SMC 2018, SCAN-B, and METABRIC) confirm its transportability across different profiling technologies. The prognostic Cox risk score (CRS) demonstrates the clinical value of the signature for overall survival validation. Furthermore, the Weighted Projected Residual Composite Uniqueness Score (WPR-CUS) and regularized Mahalanobis distance provide complementary topological and covariance-adjusted models to identify transcriptomic outliers and independent prognostic risk factors at the individual patient level, supporting a dual diagnostic-prognostic paradigm for clinical precision oncology.

8. Data and Code Availability Statement

To ensure full transparency, scientific reproducibility, and open-science compliance in accordance with journal standards (e.g., Nature Communications and Bioinformatics guidelines), all source code, software pipelines, trained models, and publicly available datasets utilized in this study are made accessible as detailed below.

8.1 Code Availability

- **Primary Source Code Repository:** The complete codebase for OncoResolve (v3.4.0), including data preprocessing, fold-contained Anti-Leakage Protocol (ALP), machine learning training routines (Linear SVM, LightGBM), Weighted Projected Residual Composite Uniqueness Score (WPR-CUS) Ridge LOO reconstruction modules, and SHAP explainability notebooks, is publicly hosted on GitHub at: <https://github.com/shubhamkja369/OncoResolve-Breast-Cancer-Transcriptomics>
- **Permanent Archival DOI:** The exact release version of the code and analysis pipeline used in this manuscript is archived on Zenodo under permanent DOI: <https://doi.org/10.5281/zenodo.21967841>.
- **Software License:** The codebase is distributed under the OSI-approved MIT License, permitting unrestricted academic and commercial re-use with proper attribution.
- **Compute Environment:** All scripts were executed in Python 3.10+ utilizing open-source libraries: scikit-learn (v1.3.0), lightgbm (v4.1.0), lifelines (v0.27.8), shap (v0.42.1), networkx (v3.1), scipy (v1.11.2), and joblib (v1.3.2).

8.2 Data Availability

All transcriptomic count matrices, microarray gene expression matrices, clinical outcome annotations, and CRISPR dependency datasets analyzed during this study are derived from open-access public repositories:

1. **TCGA-BRCA Pan-Cancer Atlas (Discovery & Internal Holdout):** Primary solid tumor transcriptomes ($N = 981$) and clinical annotations were retrieved via the NCI Genomic Data Commons (GDC) portal and cBioPortal for Cancer Genomics (brca_tcga_pan_can_atlas_2018): https://www.cbioportal.org/study/summary?id=brca_tcga_pan_can_atlas_2018
2. **SMC 2018 (External RNA-seq Validation):** RNA-seq profiles and clinical PAM50 annotations for 168 breast cancer patients from Samsung Medical Center were accessed via cBioPortal (brca_smc_2018): https://www.cbioportal.org/study/summary?id=brca_smc_2018
3. **SCAN-B / GSE96058 (External RNA-seq Validation):** RNA sequencing profiles ($N = 340$) from the Swedish Cancerome Analysis Network - Breast cohort were downloaded from the NCBI Gene Expression Omnibus (GEO) database under Accession Number GSE96058: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE96058>

4. **METABRIC (Cross-Platform Microarray Validation):** Microarray expression profiles and clinical outcome data ($N = 1,756$) were accessed via cBioPortal (brca_metabric) and the European Genome-phenome Archive (EGA) under Accession ID EGAS00001000175: https://www.cbioportal.org/study/summary?id=brca_metabric
5. **Broad Institute DepMap CRISPR Essentiality:** Functional gene essentiality dependency scores (CRISPR-Cas9 Achilles dataset, Release 23Q2) were obtained from the DepMap portal: <https://depmap.org/portal/download/>

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